

SARGVISION TRAFFIC COPILOT

Product Requirements Document (PRD)

Product: SARGVISION Traffic Copilot

Version: 1.0

Status: Proposed MVP

Pilot City: Siliguri, West Bengal

Primary Organization: SARGVISION Intelligence

Primary Users: Siliguri Traffic Police and Traffic Operations Personnel

Product Category: AI-Powered Traffic Operations Intelligence

Deployment Model: Cloud-Native, Software-First

Target MVP Timeline: 8 Weeks

1. Product Overview

1.1 Product Name

SARGVISION Traffic Copilot

1.2 Product Vision

SARGVISION Traffic Copilot is an AI-powered traffic operations intelligence platform designed to help traffic authorities understand traffic conditions across selected urban corridors.

The system combines:

- traffic-aware route information;
- continuous travel-time observations;
- historical traffic patterns;
- anomaly detection;
- corridor intelligence;
- bottleneck analysis;
- Agentic AI.

The primary objective is to answer:

What requires attention, why does it require attention, and what should the officer investigate next?

2. Problem Statement

Traffic management teams frequently operate with fragmented information.

Existing sources may include:

CCTV

Field Officers

Manual Reports

Google Maps

Phone Calls

WhatsApp Groups

Public Complaints

These sources are useful individually.

However, they create several challenges.

2.1 Lack of Unified Situational Awareness

Officers may need to manually determine:

- which roads are congested;
- whether congestion is unusual;
- which congestion is most important;
- where problems are spreading.

2.2 Lack of Historical Context

Raw traffic information does not answer:

Is this traffic condition expected?

Example:

Current Travel Time:
25 minutes

This number alone has limited meaning.

However:

Current Travel Time:
25 minutes

Expected Travel Time:
14 minutes

Deviation:
+78%

This is operational intelligence.

2.3 Reactive Operations

Traffic problems are often addressed after they become visible.

The proposed system aims to identify:

Unexpected Pattern

↓

Early Detection

↓

Priority Alert

↓

Human Investigation

2.4 Limited Human Attention

Traffic personnel cannot continuously monitor every corridor.

Therefore, the system should optimize for:

Attention Prioritization

The objective is not to display all available information.

The objective is to surface the information most likely to require attention.

3. Product Goals

Primary Goals

Goal 1

Provide a real-time and historical traffic intelligence view for selected Siliguri corridors.

Goal 2

Detect abnormal traffic patterns.

Goal 3

Identify recurring congestion and probable bottleneck zones.

Goal 4

Provide an AI Copilot interface for traffic investigation.

Goal 5

Generate daily operational traffic intelligence.

4. Non-Goals for MVP

The following capabilities are explicitly excluded from Version 1.

Autonomous Traffic Signal Control

Signal Controller Integration

Automatic Signal Timing Changes

Digital Twin

Reinforcement Learning Traffic Control

City-Wide CCTV Processing

Automated Traffic Enforcement

Facial Recognition

Vehicle Identification

License Plate Recognition

These exclusions are intentional.

The MVP must remain:

- low-cost;
- low-risk;
- software-first;
- fast to deploy.

5. Target Users

Persona 1 — Commissioner / Senior Traffic Leadership

Needs

- city-level overview;
- priority problems;
- historical trends;
- daily briefing;
- strategic insights.

Typical Questions

What are the most problematic corridors?

Has congestion improved this month?

Which locations require intervention?

Persona 2 — Traffic Control Room Officer

Needs

- current situation;
- priority alerts;
- unusual congestion;
- corridor comparison.

Typical Questions

What requires attention now?

Which corridor is deteriorating?

Is this congestion normal?

Persona 3 — Traffic Analyst

Needs

- historical data;
- pattern analysis;
- congestion trends;
- reliability analysis.

Typical Questions

Compare traffic on Fridays versus Mondays.

What are the recurring evening bottlenecks?

6. Strategic Product Modules

The platform consists of five conceptual intelligence products.

The MVP will implement three.

Module A — Traffic Incident Intelligence

MVP Status

Included.

Objective

Detect traffic conditions that significantly deviate from expected historical behaviour.

Inputs

- current travel time;
 - historical travel time;
 - route metadata;
 - time;
 - day of week.
-

Outputs

Anomaly Score

Severity

Affected Corridor

Deviation Percentage

Start Time

Duration

Confidence

Example

CORRIDOR:
Sevoke Road

CURRENT:
28 minutes

EXPECTED:
16 minutes

DEVIATION:
+75%

ANOMALY SCORE:
0.91

SEVERITY:
HIGH

Module B — Bottleneck Intelligence

MVP Status

Included.

Objective

Identify recurring congestion zones and probable traffic propagation patterns.

Inputs

- corridor travel time;
- adjacent corridor data;
- historical observations;
- temporal correlations.

Outputs

Probable Bottleneck Zone

Recurring Time Window

Affected Corridors

Propagation Pattern

Confidence Score

Module C — Mixed Traffic Intelligence

MVP Status

Future Phase.

Objective

Use computer vision to understand physical traffic behaviour.

Future Detection Types

- vehicle classification;
- Toto clustering;
- bus obstruction;
- illegal parking;
- lane blockage;
- wrong-way movement.

Module D — Traffic Operations Copilot

MVP Status

Included.

Objective

Provide natural language access to traffic intelligence.

Module E — Mobility Decision Intelligence

MVP Status

Future Phase.

Objective

Support strategic mobility and infrastructure decisions.

7. MVP Scope

The initial deployment should monitor:

10–15 Strategic Traffic Corridors

Each corridor will consist of:

```
Origin
  ↓
Destination
  ↓
Optional Intermediate Waypoints
```

8. Corridor Configuration

Example conceptual structure:

```
{
  "corridor_id": "SIL_001",
  "name": "Sevoke Road Corridor",
  "origin": {
    "latitude": 26. [xxxx],
    "longitude": 88. [xxxx]
  },
  "destination": {
    "latitude": 26. [xxxx],
    "longitude": 88. [xxxx]
  }
}
```

```
},  
"priority": "CRITICAL",  
"monitoring_interval_minutes": 5  
}
```

Final corridor definitions must be validated with Siliguri Traffic Police.

9. Functional Requirements

FR-01 — Traffic Data Collection

The system shall collect traffic-aware route information for configured corridors.

Data Fields

```
timestamp  
  
corridor_id  
  
origin  
  
destination  
  
distance  
  
normal_duration  
  
traffic_duration  
  
delay_duration  
  
delay_percentage
```

Formula

```
delay_duration  
  
=  
  
traffic_duration  
  
-
```

```
normal_duration
```

```
delay_percentage
```

```
=
```

```
delay_duration
```

```
/
```

```
normal_duration
```

```
× 100
```

FR-02 — Adaptive Monitoring

The system shall support different monitoring frequencies.

Example:

```
CRITICAL CORRIDOR
```

```
Every 5 minutes
```

```
HIGH PRIORITY
```

```
Every 10 minutes
```

```
NORMAL PRIORITY
```

```
Every 20-30 minutes
```

Monitoring frequency may increase when an anomaly is detected.

Example:

```
NORMAL STATE
```

```
15 minute polling
```

```
↓
```

```
ANOMALY DETECTED
```

```
↓
```

```
5 minute polling
```

This is required for API cost optimization.

FR-03 — Historical Traffic Baseline

The system shall calculate expected traffic conditions.

Baselines should consider:

Time of Day
Day of Week
Weekend / Weekday
Historical Travel Times

Example:

Monday
5:00 PM

Historical Median:
18 minutes

Historical Range:
16-22 minutes

Recommended Baseline Algorithm

Initial MVP:

Rolling Median
+
Percentile Bands

Example:

Expected:
Median

Normal Range:
P25 - P75

High Congestion:
Above P90

More advanced forecasting should not be required initially.

FR-04 — Traffic Anomaly Detection

The system shall identify unexpected deviations.

Proposed Anomaly Score

Conceptually:

```
Anomaly Score  
  
=  
  
Current Deviation  
+  
Historical Variance  
+  
Persistence  
+  
Spatial Spread
```

Example factors:

Current Deviation

How much current travel time differs from baseline.

Persistence

How long abnormal conditions continue.

Spatial Spread

Whether congestion spreads to connected corridors.

Severity Levels

LOW

MODERATE

HIGH

CRITICAL

Example thresholds should initially be configurable.

FR-05 — Corridor Reliability Score

The system shall calculate corridor reliability.

Proposed factors:

Travel Time Variability

+

Frequency of Severe Congestion

+

Average Delay

+

Peak Duration

=

Reliability Score

Suggested scale:

0 - 100

Where:

90-100

Highly Reliable

70-89
Reliable

50-69
Moderately Unreliable

Below 50
Highly Unreliable

The exact formula must be validated during the pilot.

FR-06 — Bottleneck Analysis

The system shall analyze temporal relationships between connected corridors.

The goal is to identify:

Congestion Origin

↓

Propagation

↓

Affected Network

Initial Methodology

For MVP:

Time-Series Correlation

+

Lag Analysis

+

Repeated Pattern Detection

Example:

Corridor A slowdown

↓

15 minutes later

Corridor B slowdown

↓

10 minutes later

Corridor C slowdown

Repeated observations may indicate a propagation pattern.

Important Constraint

The system must describe results as:

Probable bottleneck zone

unless validated by physical or CCTV evidence.

FR-07 — Traffic Pulse Dashboard

The system shall provide a city-level overview.

Display:

Current Traffic Status

Critical Corridors

High Congestion Corridors

Normal Corridors

Highest Delay

Top Priority Areas

FR-08 — Priority Alerts

The system shall generate alerts when:

- anomaly score exceeds threshold;
- severe congestion persists;
- congestion spreads;
- corridor reliability deteriorates.

Alert Schema

```
{
  "alert_id": "ALT_001",
  "corridor_id": "SIL_001",
  "severity": "HIGH",
  "type": "TRAFFIC_ANOMALY",
  "detected_at": "timestamp",
  "deviation_percentage": 65,
  "confidence": 0.91,
  "status": "ACTIVE"
}
```

FR-09 — Daily Traffic Brief

The system shall generate a daily summary.

The report should include:

Yesterday's Summary

Top Congestion Events

Most Affected Corridors

Recurring Patterns

Today's Expected Risk

Recommended Monitoring Areas

FR-10 — AI Traffic Copilot

The system shall provide a conversational interface.

The Copilot must support:

Current Status Queries

Historical Queries

Comparison Queries

Anomaly Investigation

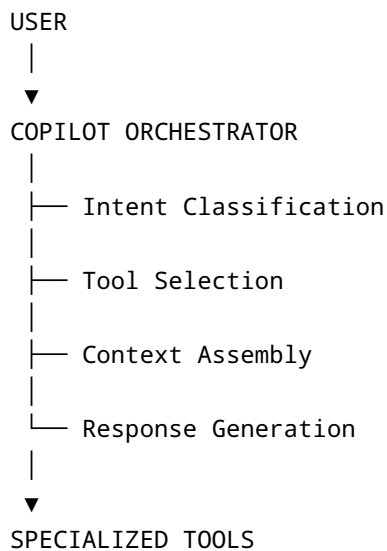
Corridor Analysis

Daily Brief Queries

10. AI Copilot Architecture

The system should not use uncontrolled autonomous agents.

Recommended architecture:



11. Copilot Tools

Tool 1

get_current_traffic_status()

Inputs:

```
corridor_id  
time
```

Outputs:

```
current_travel_time  
normal_travel_time  
delay  
severity
```

Tool 2

analyze_historical_pattern()

Inputs:

```
corridor_id  
  
date_range  
  
time_range
```

Outputs:

```
average  
  
median  
  
variance  
  
peak_period  
  
trend
```

Tool 3

detect_traffic_anomaly()

Outputs:

```
anomaly_score  
  
severity  
  
duration  
  
confidence
```

Tool 4

compare_corridors()

Inputs:

```
corridor_list  
  
time_range
```

Outputs:

```
ranking  
  
delay  
  
reliability
```

Tool 5

investigate_bottleneck()

Inputs:

```
corridor_id  
  
time_window
```

Outputs:

```
probable_origin  
affected_corridors  
propagation_pattern  
confidence
```

Tool 6

generate_daily_brief()

Outputs:

```
summary  
alerts  
risk_areas  
patterns  
recommendations
```

12. Agentic AI Design Principles

The Copilot must follow strict principles.

Principle 1

The LLM does not calculate traffic metrics.

Analytics services calculate metrics.

Principle 2

The LLM does not invent observations.

Every factual claim should be supported by:

- database data;
- analytics output;
- external validated source.

Principle 3

The LLM explains.

Analytics calculates.

```
graph TD;
    A[ANALYTICS ENGINE] --> B[Structured Evidence];
    B --> C[LLM];
    C --> D[Human Explanation];
```

Principle 4

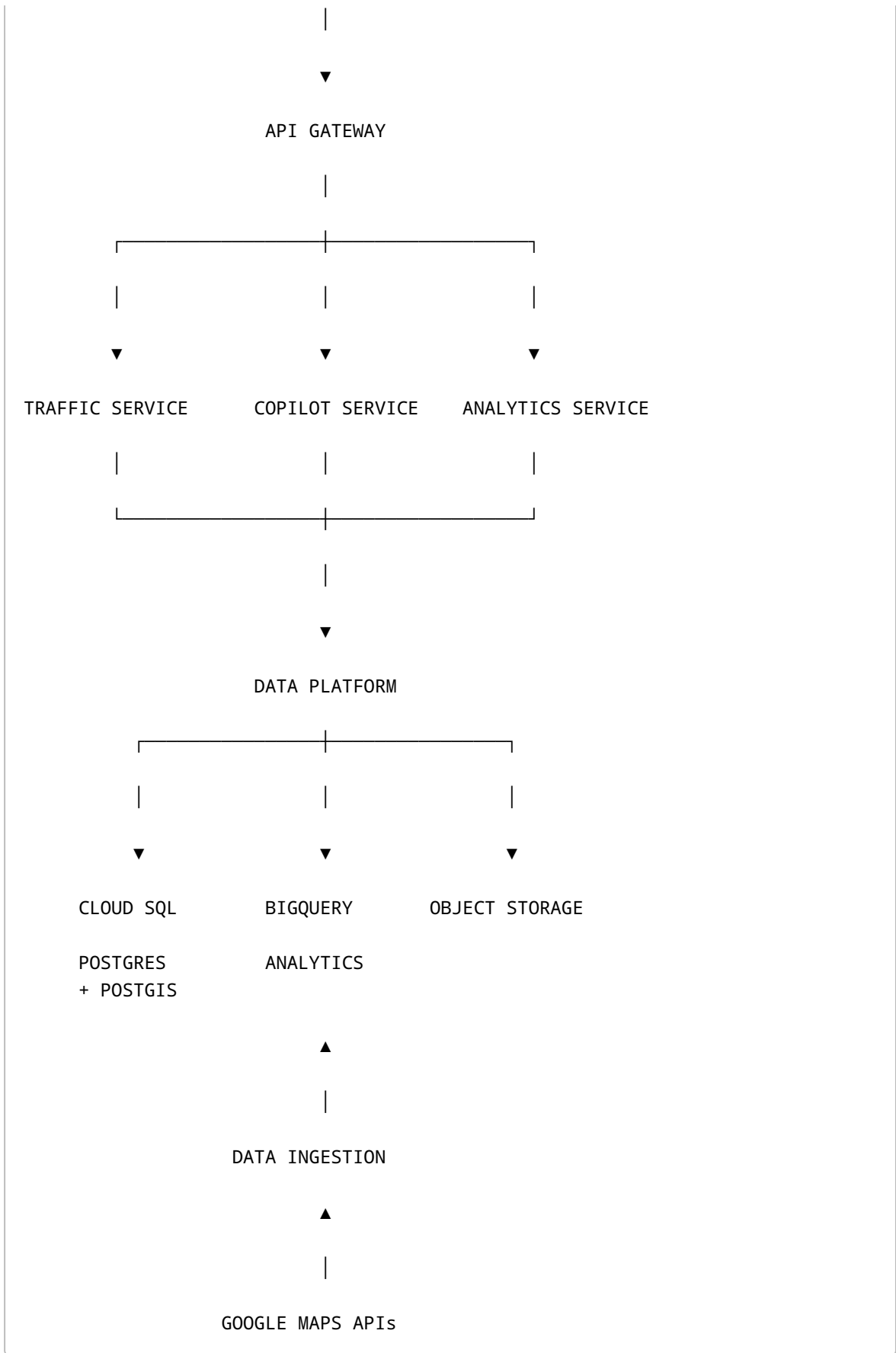
Recommendations must be clearly categorized.

```
graph TD;
    A[OBSERVATION] --> B[INFERENCE];
    B --> C[RECOMMENDED INVESTIGATION];
```

These categories should never be mixed.

13. System Architecture

```
graph TD;
    A[USERS] --> B[NEXT.JS FRONTEND];
```



14. Recommended Technology Stack

Frontend

Next.js
TypeScript
Tailwind CSS
Map Visualization

Backend

Python
FastAPI
Pydantic

Agent Framework

Recommended:

Google ADK

Alternative abstraction layer may be used if required.

LLM

Gemini via Vertex AI

Infrastructure

Google Cloud Run

Scheduling

Cloud Scheduler

Async Processing

Cloud Tasks
or
Pub/Sub

Transactional Database

Cloud SQL

PostgreSQL

PostGIS

Analytics

BigQuery

Optional initially depending on data volume.

Monitoring

Cloud Logging

Cloud Monitoring

15. Data Architecture

Core Tables

corridors

```
corridor_id  
  
name  
  
origin  
  
destination  
  
priority  
  
monitoring_frequency  
  
status
```

traffic_observations

```
observation_id  
  
timestamp  
  
corridor_id  
  
normal_duration  
  
traffic_duration  
  
delay_duration  
  
delay_percentage  
  
distance  
  
source
```

traffic_baselines

```
baseline_id
```

```
corridor_id  
  
day_type  
  
time_window  
  
median_duration  
  
p25_duration  
  
p75_duration  
  
p90_duration
```

anomalies

```
anomaly_id  
  
corridor_id  
  
timestamp  
  
anomaly_score  
  
severity  
  
deviation  
  
duration  
  
status
```

bottleneck_patterns

```
pattern_id  
  
primary_corridor  
  
affected_corridor  
  
time_lag  
  
correlation  
  
frequency
```

confidence

copilot_interactions

interaction_id

user_id

timestamp

query

tools_called

response

evidence_reference

16. Data Collection Strategy

Continuous API polling must be cost-controlled.

Recommended approach:

CRITICAL CORRIDORS

Peak Hours:
Every 5 minutes

Off-Peak:
Every 15 minutes

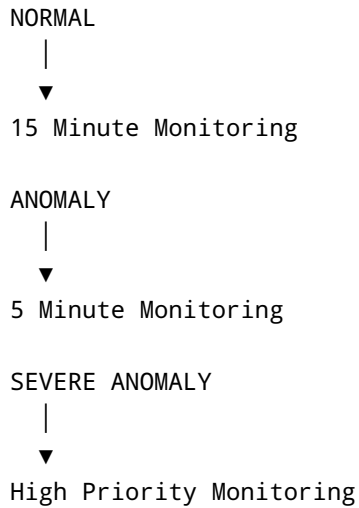
STANDARD CORRIDORS

Peak Hours:
Every 10 minutes

Off-Peak:
Every 30 minutes

17. Dynamic Monitoring Strategy

The monitoring engine should adapt.



This improves:

- operational awareness;
- API efficiency;
- cloud cost.

18. Analytics Engine

The Analytics Engine consists of four core components.

18.1 Baseline Engine

Calculates expected traffic.

Inputs:

Historical Travel Time

Time of Day

Day Type

The inputs for the Baseline Engine are listed in a box: Historical Travel Time, Time of Day, and Day Type.

18.2 Anomaly Engine

Calculates:

Current Observation

vs

Expected Range

18.3 Pattern Engine

Detects:

Recurring Peaks

Recurring Congestion

Day-of-Week Patterns

Travel-Time Trends

18.4 Propagation Engine

Detects potential relationships between corridors.

Example:

A

↓

B

↓

C

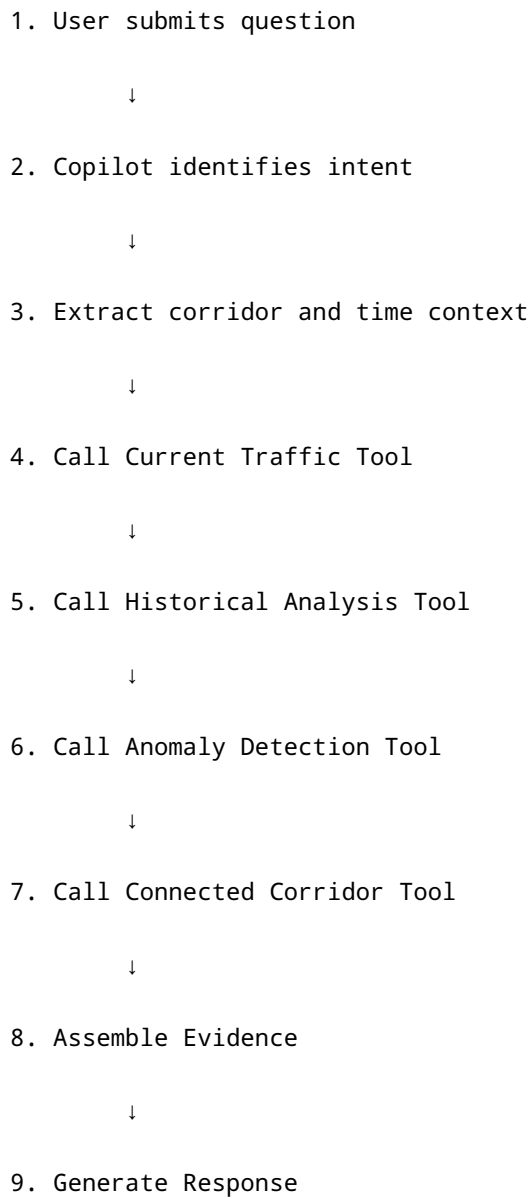
If repeated time-series relationships are detected, the system records a probable propagation pattern.

19. Copilot Query Flow

Example query:

Why is Sevoke Road congested?

System workflow:



20. Response Format

Copilot responses should be structured.

Example:

Current Observation

Current travel time is 24 minutes.

Expected travel time for this period is approximately 15 minutes.

Historical Context

Current traffic is 60% above the historical median.

Pattern

The congestion began approximately 30 minutes earlier than the normal evening peak.

Impact

Two connected corridors are also experiencing elevated delays.

Recommended Investigation

Physical verification may be required near the probable bottleneck zone.

This format prevents vague AI responses.

21. Confidence Framework

All intelligence outputs should contain confidence.

Example:

HIGH

MEDIUM

LOW

Confidence Sources

Amount of Historical Data

Pattern Consistency

Signal Strength

Data Quality

Spatial Correlation

22. Security Requirements

The MVP should implement:

- authenticated access;
- role-based access;
- audit logging;
- API key management;
- encrypted storage;
- encrypted communication.

23. Privacy Requirements

The initial MVP should avoid collecting:

Faces

Vehicle Number Plates

Personal Identity Information

The initial data model focuses on:

- aggregated travel times;
- corridor performance;
- traffic patterns.

Future CCTV modules must undergo separate privacy and governance review.

24. Performance Requirements

Dashboard

Target:

Page Load
< 3 seconds

Copilot

Target:

Simple Query
< 5 seconds

Complex Investigation
< 15 seconds

Alert Detection

Target:

Within one monitoring interval

25. MVP Success Metrics

Data Metrics

Data Collection Success Rate

API Failure Rate

Observation Completeness

Analytics Metrics

Anomaly Detection Precision

False Positive Rate

Pattern Stability

User Metrics

Daily Active Officers

Copilot Queries

Alert Review Rate

Officer Feedback

Operational Metrics

Number of Significant Events Detected

Time Saved in Investigation

Recurring Bottlenecks Identified

Priority Areas Investigated

26. MVP Evaluation Framework

The pilot should not be judged based on:

Did AI solve all traffic problems?

Instead evaluate:

Question 1

Did the system identify traffic situations officers considered relevant?

Question 2

Did historical context improve decision-making?

Question 3

Did anomaly alerts identify unusual traffic conditions?

Question 4

Did the Copilot reduce investigation effort?

Question 5

Did the platform identify recurring patterns that were previously undocumented?

27. Product Roadmap

Phase 1 — MVP

Duration

8 weeks.

Features

Traffic Data Collection

Historical Database

Traffic Pulse

Anomaly Detection

Corridor Analytics

Bottleneck Analysis

Daily Brief

Traffic Copilot

Phase 2 — Operational Context

Add:

Road Closures

Manual Incident Input

Police Reports

Events

Festival Calendar

Phase 3 — CCTV Intelligence

Pilot:

2-3 Strategic Locations

Capabilities:

Vehicle Classification

Queue Detection

Parking Detection

Toto Clustering

Road Obstruction Detection

Phase 4 — Root Cause Intelligence

Combine:

Traffic Data

CCTV

Officer Reports

Historical Patterns

The AI can then generate stronger evidence-based investigations.

Phase 5 — Mobility Decision Intelligence

Support:

Infrastructure Prioritization

Bottleneck Intervention

Corridor Improvement Analysis

28. Engineering Backlog

Epic 1 — Corridor Management

Tasks

- Create corridor data model
 - Corridor CRUD APIs
 - Map configuration interface
 - Priority configuration
-

Epic 2 — Traffic Data Collection

Tasks

- Google Maps API integration
 - Route request service
 - Route Matrix integration
 - Scheduler
 - Retry mechanism
 - Rate limiting
 - Cost monitoring
-

Epic 3 — Historical Analytics

Tasks

- Observation storage
 - Baseline calculation
 - Time-series aggregation
 - Percentile calculations
-

Epic 4 — Anomaly Detection

Tasks

- Deviation calculation
- Severity scoring
- Persistence analysis
- Alert generation

Epic 5 — Bottleneck Intelligence

Tasks

- Corridor relationship model
 - Time-lag analysis
 - Correlation engine
 - Pattern persistence scoring
-

Epic 6 — Dashboard

Tasks

- Traffic Pulse
 - Map
 - Corridor list
 - Alert interface
 - Historical charts
-

Epic 7 — AI Copilot

Tasks

- Intent recognition
 - Tool definitions
 - Agent orchestration
 - Evidence aggregation
 - Response formatting
 - Guardrails
-

Epic 8 — Daily Brief

Tasks

- Summary engine
 - Risk ranking
 - Historical comparison
 - Automated generation
-

29. Eight-Week Implementation Plan

Week 1

Foundation

- Finalize corridors
 - Configure GCP
 - Configure database
 - Google Maps integration
-

Week 2

Data Collection

- Scheduler
 - Route collection
 - Data persistence
 - Monitoring
-

Week 3

Historical Intelligence

- Baseline engine
 - Aggregations
 - Initial pattern analysis
-

Week 4

Anomaly Detection

- Anomaly scoring
 - Severity classification
 - Alert generation
-

Week 5

Dashboard

- Traffic Pulse
 - Corridor analysis
 - Alert interface
-

Week 6

Bottleneck Intelligence

- Corridor relationship model
 - Lag analysis
 - Pattern visualization
-

Week 7

AI Copilot

- Tool integration
 - Gemini
 - Agent orchestration
 - Evidence-based responses
-

Week 8

Pilot Validation

- Testing
 - Historical validation
 - Officer feedback
 - Dashboard refinement
 - Demonstration preparation
-

30. Risks

Risk 1 — Limited Historical Data

Mitigation

Start collecting immediately.

Use:

Pilot Data

+

Short-Term Baselines

+

Risk 2 — Google API Cost

Mitigation

Use:

- adaptive polling;
 - corridor prioritization;
 - peak-hour monitoring;
 - caching;
 - dynamic sampling.
-

Risk 3 — False Positives

Mitigation

Use:

- confidence scores;
 - persistence requirements;
 - human validation.
-

Risk 4 — AI Hallucination

Mitigation

Strict tool-based architecture.

The AI cannot generate quantitative claims without data.

Risk 5 — User Adoption

Mitigation

Focus on one simple question:

What requires my attention?

Avoid building a complex dashboard.

31. Product Principles

Principle 1

Human authority remains central.

Principle 2

AI provides intelligence, not uncontrolled decisions.

Principle 3

Every important insight should be evidence-based.

Principle 4

Start without expensive infrastructure.

Principle 5

Build progressively.

```
DATA
↓
INSIGHT
↓
TRUST
↓
ACCESS
↓
DEEPER INTELLIGENCE
```

32. Final Product Definition

SARGVISION Traffic Copilot

An AI-powered traffic intelligence system that combines traffic data, historical analysis and Agentic AI to help traffic authorities:

Know what is happening.

Understand what is unusual.

Identify recurring bottlenecks.

Prioritize attention.

Investigate traffic conditions through natural language.

33. MVP Success Definition

The MVP will be considered successful if a Traffic Officer can open the platform and receive a meaningful answer to:

What requires my attention today?

And then ask:

Why?

The system should provide an evidence-based explanation using available traffic data.

That is the first step toward a much larger vision:

Building AI-powered urban mobility intelligence for Indian cities.

Final Product Promise

Know what is happening.

Understand what is unusual.

Investigate intelligently.

Help humans make better operational decisions.

SARGVISION TRAFFIC COPILOT

AI-Powered Traffic Intelligence for Siliguri